# Awokola's Law: A Heuristic Decay Profile for Context Window Attention Loss in Large Language Models

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**Abstract**

This paper presents a phenomenological and heuristic framework for modeling information retrieval and attention degradation within the context windows of causal autoregressive language models. We formalize the empirical "lost in the middle" phenomenon as a continuous negative exponential distribution, mapping retention probability against token position, parameter scale, and textual distraction factors.

## 1 Introduction

Recent empirical evaluations of massive sequence processing in large language models (LLMs) indicate a non-linear degradation of retrieval accuracy. Specifically, models exhibit high performance at the boundaries of the input sequence (primacy and recency effects) while demonstrating a significant performance drop near the geometric midpoint. This paper proposes a unified performance metric, known formally as the *Awokola Attention Dynamics Law*, to mathematically abstract this behavior.

## 2 The Mathematical Formulation

The attention retrieval probability $A$ of a target asset embedded within a sequence is governed by the continuous negative exponential distribution:

$$A(x, P, D) = \exp\left(-\frac{\gamma}{D \cdot \ln(P) \cdot [1 + 4(x - 0.5)^2]}\right) \tag{1}$$

Where the system variables and bounds are defined as follows:

- $A \in [0, 1]$: Attention Retrieval Probability. The mathematical likelihood of successful token synthesis and extraction during generation.
- $x \in [0, 1]$: Normalized Token Position. The relative coordinate of information location, where $x = 0$ represents the prompt head and $x = 0.5$ represents the absolute sequence midpoint.
- $P \in \mathbb{R}^+$: Model Parameter Scale. Representing total dense computational capacity.
- $D \in \mathbb{R}^+$: Context Quality & Formatting Density. A variable scaling factor representing information entropy and text formatting noise.
- $\gamma \in \mathbb{R}^+$: Systemic Architectural Fatigue Constant. An intrinsic factor mapping transformer head scaling and positional embedding overhead.

# 3 Boundary Analysis & Theoretical Proofs

## 3.1 The Valley of Forgetfulness

When a target asset is located precisely at the geometric midpoint ($x = 0.5$), the position-dependent Lorentzian term $4(x - 0.5)^2$ collapses to 0. The denominator achieves its global minimum, maximizing the magnitude of the negative exponent:

$$A_{\min} = \exp\left(-\frac{\gamma}{D \cdot \ln(P)}\right) \tag{2}$$

This establishes the operational performance floor of a given network architecture, proving why information retrieval performance hits a profound trough at the center of large text sequences.

## 3.2 Infinite Parameter Scaling Limit

As parameter scale $P \to \infty$ or context noise approaches a perfectly clean baseline ($D \to 0$), the denominator approaches infinity, driving the exponential term asymptotically toward zero:

$$\lim_{P \to \infty} A(x, P, D) = \exp(0) = 1 \tag{3}$$

This mathematically proves that extreme model scaling suppresses positional forgetfulness, flattening the retrieval curve into an ideal performance ceiling.